

Assignment#3

Class: BSAI-B, BCSB

Subject: Linear Algebra

Date: 5/6/26

Question 1

A renewable energy company operates three energy stations: **Solar**, **Wind**, and **Hydro**. During each distribution cycle, energy is transferred among the stations according to the matrix

$$A = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 1 & 2 \end{bmatrix}$$

To identify the dominant energy flow patterns and simplify long-term forecasting, the company's analysts need to study the matrix.

Questions:

1. Find the eigenvalues of the matrix A.
2. Find the corresponding eigenvectors for each eigenvalue.
3. Determine whether the matrix A is diagonalizable.
4. If it is diagonalizable, find matrices P and D such that $A = PDP^{-1}$.

Question#2

A data center monitors the flow of information among three servers: **Server A**, **Server B**, and **Server C**. The traffic dynamics are represented by

$$A = \begin{bmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

The network administrator wants to identify the principal traffic modes and simplify future calculations.

Questions:

1. Find the eigenvalues of the matrix A.
2. Find the corresponding eigenvectors for each eigenvalue.
3. Determine whether the matrix A is diagonalizable.
4. If it is diagonalizable, find matrices P and D such that $A = PDP^{-1}$.

Question#2

A graphics designer uses a transformation to stretch and rotate objects on a computer screen. The transformation $T: \mathbb{R}^2 \rightarrow \mathbb{R}^2$ is defined by

$$T(x, y) = (2x + y, x + 2y)$$

Verify whether T is a linear transformation.

